Fiscal Year:	FY 2025	**Task Last Updated:**	FY 09/30/2024
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Project Title:	Thermal Fluctuations of Colloidal Gels		
Division Name:	Physical Sciences		
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Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
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Space Biology Element:	None		
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Comments:			
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
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Task Description:	We propose to study colloidal gels using the data from the Advanced Colloids Experiment-Microscopy-1 (ACE-M-1) experiments. Colloidal gels are formed from sticky micron-sized solid particles in a liquid, where over time the particles stick together and form random clumps that eventually bridge across the entire sample chamber in tangled networks. The ACE-M-1 data set consists of a large number of optical microscopy movies of colloidal gels. In microgravity, these gels are long-lived, rather than collapsing under their own weight as happens in ground-based experiments. Our group has expertise in analyzing microscope images, using particle tracking, particle image velocimetry, and differential dynamic microscopy. The movies reveal the gels have visible thermal fluctuations that likely reveal information about their rheological properties, local elasticity, etc. There is also some clear sense of length scales: some gels have thick aggregated regions, while others have more tenuous gel strands and free particles. We can connect structure and dynamics. We propose to analyze these movies to pull out this information, and to perform complementary ground-based experiments to validate the observations. For example, the Physical Sciences Informatics (PSI) movies are all 2D cuts through 3D samples, and we will use ground-based confocal microscopy to cross-check the samples with some 3D data sets; to confirm inferences about 3D structure from the 2D images. A particular appealing feature of the ACE-M-1 experiments is that they studied polydispersity (gels made of mixtures of two distinct particle types) and polydisperse systems are a long-time interest of the Principal Investigator (PI). We will use our ground-based experiments to study gels with higher polydispersity, which should in turn suggest new microgravity investigations in the future. This PSI investigation will be done by the Weeks lab at Emory University; there will not be any other collaborators.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Colloids are suspensions of small solid particles suspended in a liquid. Examples include paint, ink, pastes, and blood. "Small" means the particle diameters range from ~10 nm to ~10 microns. Thermal motion is relevant: Brownian motion allows particles to diffuse. Often precautions are taken to prevent the solid particles from sticking together. If particles have attractive interactions, they can stick together in free-floating aggregates, or large tendrils that can span across the system. The latter is a colloidal gel. Colloidal gels are used in applications such as water purification, skin creams, and also show up in some food products such as jellies and jams. In food, colloidal gels modify the texture and shelf-life stability of the food. Our NASA-funded study of colloidal gels should improve our understanding of long-term stability of colloidal gels, as well as how they initially form. One project we are pursuing is the study of colloidal gels made from highly polydisperse particles: particles of a wide range of sizes. This is more realistic for industrial materials, as compared to a lot of prior work which studied particles that were are nearly the same size. Thus, our work will help understand the structure of these more realistic, easier to make colloidal gels.
Task Progress:	Our project primarily involves analyzing data from NASA's Physical Sciences Informatics (PSI) database. In particular, we are analyzing microscope images of colloidal gels from the ACE-M-1 experiment (Advanced Colloids Experiment). The ACE-M-1 experiment resulted in 530 GB of images, which we successfully downloaded in 2022. Colloids are small (micron-sized) particles in a liquid. In these experiments, the particles are made to be sticky, so they stick together into a network of tendrils of particles. This is the colloidal gel state. The ACE-M-1 experiment was done on the International Space Station so that the colloidal gels could be studied in microgravity conditions. The reason microgravity is important is that, as the particles begin to stick together, they form heavy clusters that can sink in normal gravity. This ultimately limits the shelf life of a colloidal gel, but also changes the overall structure. We wish to understand what the gel structure would be like if this limit is overcome in microgravity conditions. RESULTS FOR MICROGRAVITY DATA: The ACE-M-1 data set has nine distinct experiments on 8 different samples, where the different samples are made with different levels of attractive interaction – that is, different levels of particle stickiness. Undergraduate student Swagata Datta studied the ACE-M-1 data in 2022 working with the Principal Investigator (PI) Eric Weeks, and subsequent analysis in the past year was primarily done by the PI. We have determined that of the nine experiments, four are suitable quality to be fully analyzed. These experiments have 50-60 hours of data each. Fortunately, these four experiments include the one with the highest attractive interaction, and also the one with the lowest attractive interaction, thus spanning the entire range. The samples are composed of equal amounts of small (1.8 micron diameter) and large (2.2 micron diameter) particles. Much of our analysis and results was reported in last year's annual report. We briefly reprise the key points here. First, the colloidal gel samples show aging: the dynamics slow down as a function of time. Second, the slowing of dynamics is related to particles sticking together over time. That is, even particles stuck in a gel still undergo some Brownian motion, and this can cause coarsening of the gel: particles occasionally rearrange into more stable configurations. In the past year, we have elaborated on this analysis by examining the role of freely diffusing particles. Initially, when the samples are first formed, the samples that have colloidal gels present at the earliest stages of observation also have some freely diffusing particles; this is about 10 – 40% of the small particles. Over the course of tens of hours, these particles become incorporated into the gels. Even separating out these particles from the analysis, the particles that are in the gels – whether initially or at long times – also slow down over time. This indicates that the gel partially restructures itself into more stable configurations, or perhaps that as the free particles are added to the gel, they do so in a way that makes the entire gel more rigid. Additionally, in last year's NASA Task Book Report, we noted that for the non-gel forming sample (with the least amount of depletant polymer, and thus least attractive interactions), the number of particles observed over time decreased dramatically. A conversation with Prof. John Crocker (University of Pennsylvania) shed light on the problem: he pointed out that they are most likely sticking to the sample chamber walls due to the depletion force. The depletion force is twice as large between a particle and a wall as between two particles; so while the depletion force is not sufficient to stick the particles together to form a gel, it is enough to cause particles to stick to the walls. NEW GROUND-BASED EXPERIMENTS: Two years ago we started ground-based experiments on colloidal gels, and in the past year this effort has been led by lab technician Ben Lonial. Ben earned his undergraduate degree in physics (with honors) from Emory University in May 2023. He took a gap year and spent it focusing on the NASA project. Ben made colloidal gels with particles with extreme size polydispersity.

	Our goal is to understand how the mixture of particle sizes changes the gel structure. While the ACE-M-1 data uses two different particle sizes, they are fairly similar in size (1.8 micron and 2.2 micron diameter) and the gel structure does not vary a huge amount due to these sizes. On the other hand, with the larger range of sizes in Ben's new samples, we see differences. The size distribution is roughly log-normal, with the largest particles more than 15 times the size of the smallest particles. The volume fractions studied ranged from 0.01 to 0.50; at the latter, half of the volume of the sample is solid particles and the other half the liquid solvent. Our motivating question is to understand how random the structure is: What geometric constraints influence the structure? For example, particles are not all equal. We verify that larger particles have more contacts. Surprisingly, this is not proportional to surface area. The number of neighboring particles scales as approximately R0.8 rather than R2, which would be the case if the surface area was the key factor. Additionally, samples at higher volume fractions result in all particles having more neighbors, which makes sense. But even at the lowest volume fractions (down to 0.01), nearly all particles are stuck to at least one other particle. This is because this particular colloidal sample was composed of highly attractive particles, so that if any two come into close proximity they will stick permanently. Graduate student David Meer has been doing complementary simulations of random close packed circles or spheres. One set of simulations uses the exact same size distribution as seen in Ben's experimental data. In this case, the number of nearest neighbor particles in the simulation, as a function of particle radius, is the same overall shape as the experimental colloidal gel data. This suggests that the contact numbers are set by geometry and the particle size distribution, rather than anything physically different between the different particle sizes. Ben also looked at tetrahedral structures in his data: places where four particles are mutually touching. He finds that the larger particles act as hubs for the formation of tetrahedra; but this is in large part because they generally have more neighbors, as discussed in the previous paragraph, so it is easier for them to have more mutually touching neighbors that thus form a tetrahedron. We have written up this work on highly polydisperse colloidal gels for publication, and a preprint is available at arXiv: 2406.10321. [Ed. Note: See Bibliography.] This was submitted to Phys. Rev. E, and we are now revising the manuscript in response to the referee concerns. We are also nearly finished writing up the paper analyzing the ACE-M-1 data and expect to post a preprint on arXiv soon. Another happy outcome during the course of this grant is that summer undergraduate researcher Swagata Datta is now a graduate student in mathematical physics at the University of Alberta, and undergraduate/lab technician Ben Lonial is now a graduate student in physics at the University of California/Santa Barbara.
Bibliography Type:	Description: (Last Updated: 05/04/2026)
Articles in Other Journals or Periodicals	Lonial BF, Weeks ER. "Microstructure of polydisperse colloidal gels." arXiv preprint arXiv:2406.10321. 2024 Jun 14. https://doi.org/10.48550/arXiv.2406.10321 , Jun-2024